



Prototype communications antenna embroidered into cloth

who are finding ways to incorporate radio antennae directly into clothing using a plastic film and metallic thread.

The new antenna design has a range four-times that of a conventional antenna worn on the body by American soldiers today. The researchers developed a computer control device that lets multiple antennae work together in a single piece of clothing.

The prototype antenna was created by etching thin layers of brass on a commercially available plastic film called FR-4. The film is light and flexible, and can be sewn into the fabric. They attached it into a vest at four locations—chest, back and both the shoulders. The computer controller is worn on a belt.

The developed design surrounds the body with several antennae that work together to transmit or receive a signal, no matter which way a person is facing. The integrated computer control device senses body movements and switches between the antennae to activate the one with the best performance, given the body's position.

Key to the technology was the engineers' development of network communications coding to coordinate the signals among the antennae.

Industry's first MIPI M-PHY transmitter compliance test software

Agilent Technologies has introduced

what it claims to be the industry's first MIPI M-PHY compliance test software package for transmitter validation.

The U7249A runs on the Infiniium 90000 X-Series oscilloscope, which delivers real-time bandwidth of up to 33 GHz with the only available 30GHz probing system. With low-noise and low-jitter measurement-floor performance, the scope ensures measurement accuracy.

The compliance test software provides design and validation engineers with a fast and easy way to validate and debug embedded M-PHY data links. It supports M-PHY testing for gear 1, 2 and 3 speeds of up to 5.8 Gbps, automated scope measurement setup and programming for increased accuracy and repeatability in test results, and automated margin analysis and pass/fail conformance reporting.

Power semiconductor chips on 300mm thin wafer

Infineon Technologies claims to have produced the first chips on a 300-millimetre thin wafer for power semiconductors.

Energy efficiency and energy saving are becoming the prime requirements for all electrically powered industrial or household applications. These energy-saving chips are useful for consumer electronics devices, for instance, in flat-screen TVs and gaming consoles.

The chips exhibit the same behaviour as the power semiconductors made on

200-millimetre wafers—demonstrated by application tests using metal-oxide semiconductor field-effect transistors (MOS-FETs) for high-voltage applications.

Industry's smallest IO-link transceiver

Maxim Integrated Products has introduced MAX14821—which is said to be the smallest IO-Link physical-layer (PHY) transceiver in the industry.

Encased in a 2.5x2.5mm² wafer level package, the device has an extra set of a digital output and input plus dual integrated low-dropout linear regulators that provide added flexibility with complex sensor modules. The transceiver is optimised for point-to-point communication between industrial controllers, and sensor modules and actuators. It allows calibration, diagnostics and data transmission using a standard three-conductor wire and connectors.

The MAX14821 supports all specified IO-link data rates: 4.8 kbps (COM1), 38.4 kbps (COM2) and 230.4 kbps (COM3). It also supports 24V binary sensor/actuator applications, so manufacturers can leverage the advanced performance capabilities of the device in non-IO-link applications.

Data centre simulator

Fujitsu Laboratories and Fujitsu Laboratories of Europe have developed a simulation technology that can instantly test for changes in power consumption throughout a data centre when operating status of servers or air-conditioning equipment is altered.

The technology determines thermal flows inside a building at calculation speeds over a thousand times faster than conventionally possible, and completely models the ICT hardware, air-conditioning and other equipment of a running data centre. This, in turn, enables accurate calculations of a data centre's power consumption. The technology will enable users to test the effectiveness of energy-saving measures prior to execution in a running data centre. This has the potential to significantly reduce the overall power consumption of data centres.

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